

- 5 The motor of a small boat has a propeller of diameter 0.200 m. When the boat is starting from rest, the propeller sends a stream of water backwards at a speed of 10.0 m s^{-1} . Half of the work done by the motor is transferred to the water as its kinetic energy.

What is the power of the motor, taking density of water to be 1000 kg m^{-3} ?

- A 1.25 kW
- B 6.50 kW
- C 15.7 kW
- D 31.4 kW

6 A skateboarder of mass m skates on a smooth semi-circular half-pipe as shown. He starts